Mr Dimelow

3 Processes, techniques and specialist tools

3.4 Permanent and Semi-Permanent Joining Techniques

c) heat – oxy-acetylene welding, MIG welding, brazing, hard soldering, soft soldering (including use of specialist tools)

Joining with Welding







In this presentation you will:

- explore welding processes and types of joint and weld

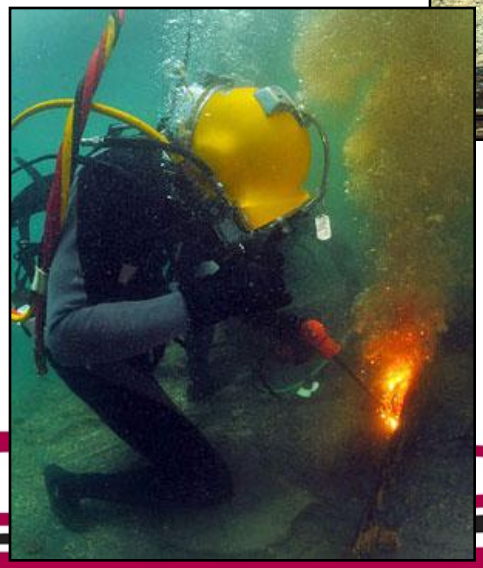
Introduction

Welding is a process for permanently joining materials, usually metals.





It is used in many industries, including automotive, building construction, marine and aerospace.



Welding can even be carried out under water.

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The Welding Process

The materials at the joint are heated to a molten state and allowed to solidify.



Most welding processes use a filler material in the joint.



Some welding processes use a combination of heat and pressure.



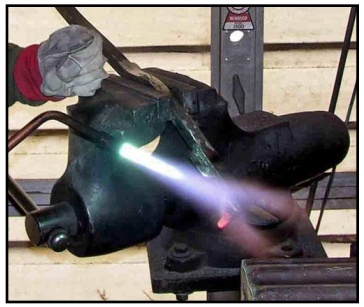
Welding Processes

There are two main types of welding processes; fusion welding and solid state welding.

In fusion welding, the base metals melt and flow together, then solidify.

In solid state welding, the materials soften at the join but do not flow together.

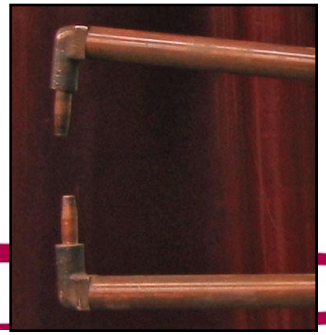
Fusion Welding



Gas welding



Electric arc welding

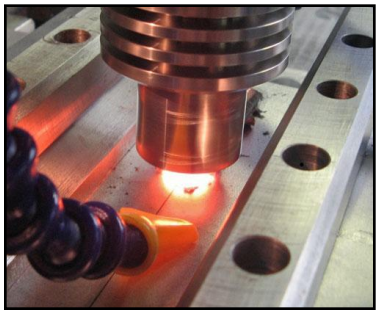


Resistance welding

Solid State Welding



Forge welding



Friction welding



Ultrasonic welding

Next >

Question 1

Which of the following is a fusion welding process?

A) Forge welding

B) Arc welding

C) Friction welding

D) Ultrasonic welding

Next >

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A) Forge welding



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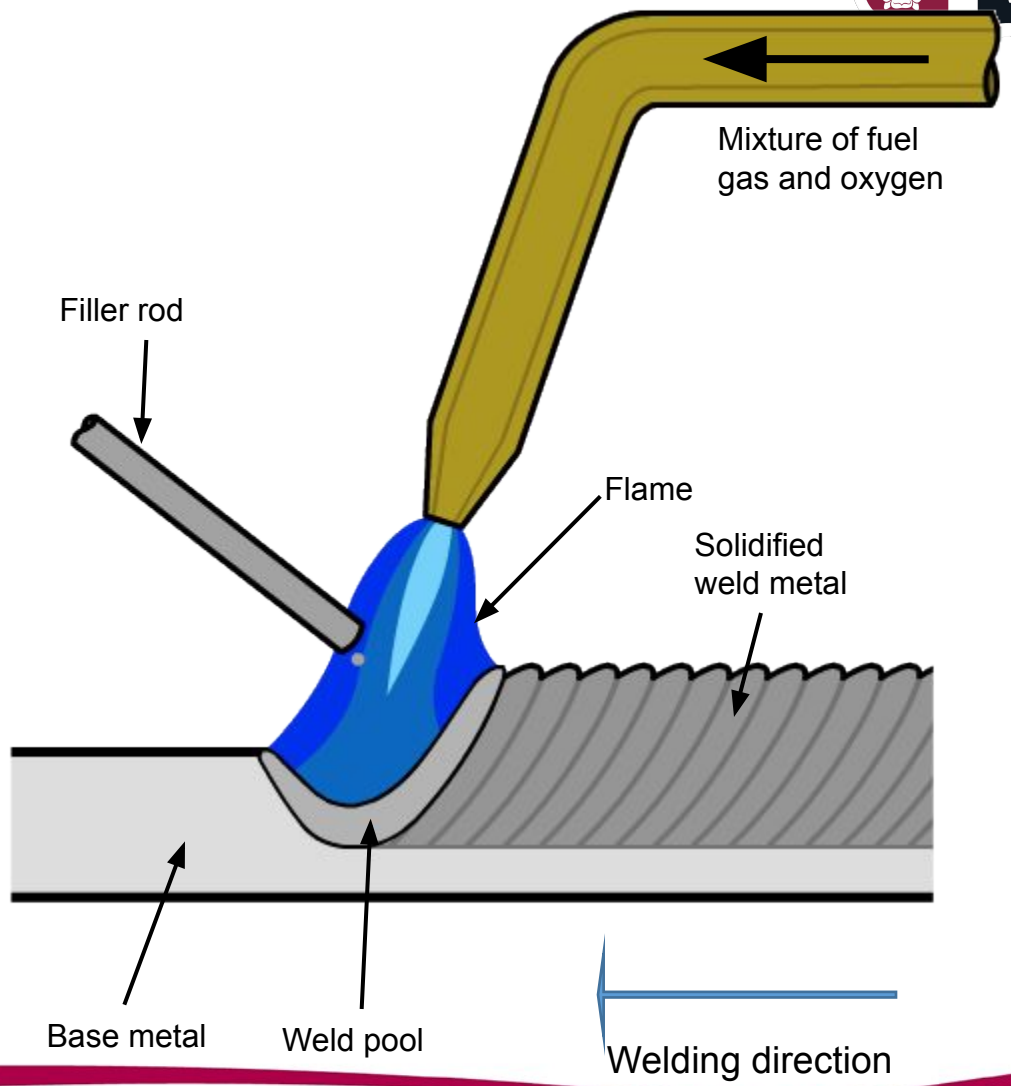
Gas Welding

During gas welding, fuel gas and oxygen burn with a high-temperature flame.

The heat melts both the base metal and a metal filler rod.

The molten metals flow together, then solidify.

The filler rod contains flux that vaporises to displace air and stop oxide contamination.



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Oxyacetylene Gas Welding



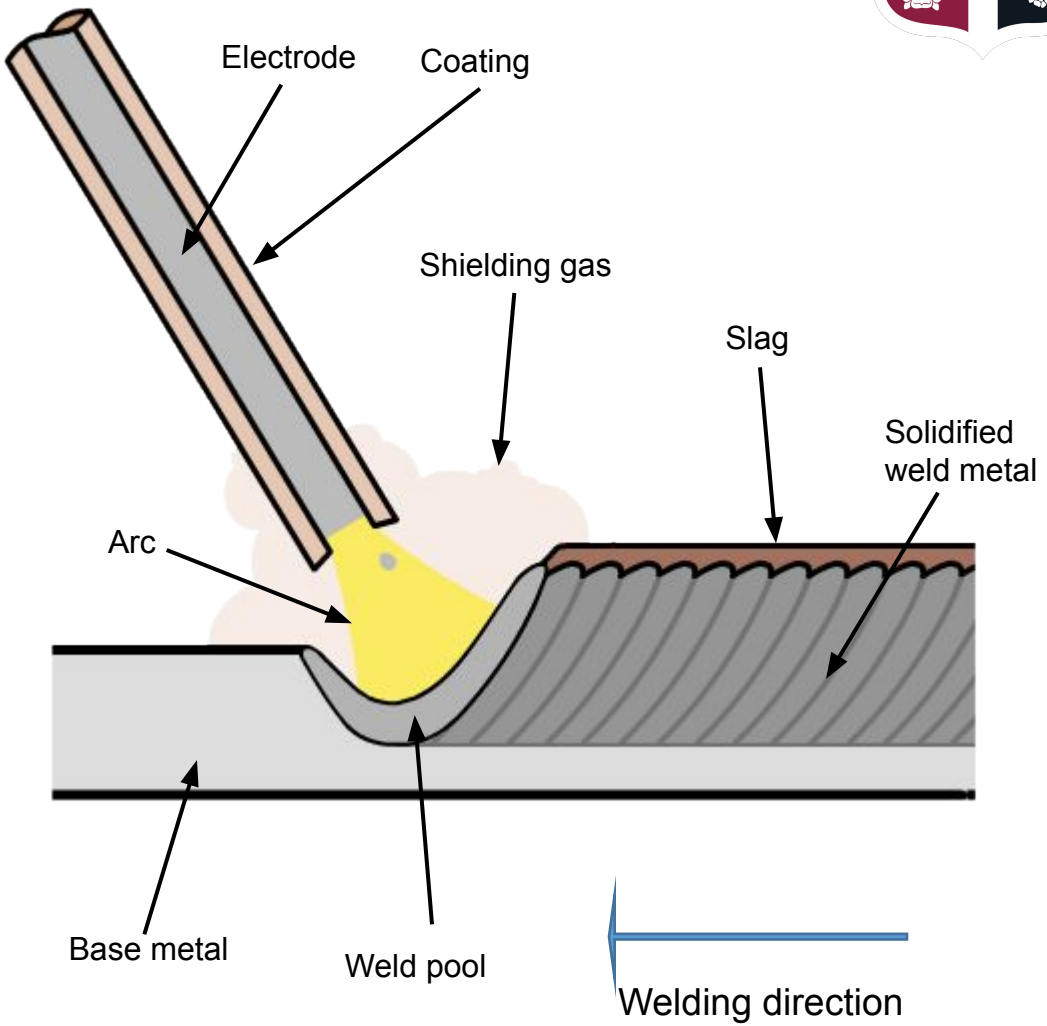
Arc Welding

Arc welding uses an electric arc to heat and melt metals.

Most arc welding processes use a consumable electrode as the filler metal.

A shielding gas is used to prevent weld oxidisation.

Some processes generate the gas by vaporising a coating on the electrode.



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Electric Arc Welding

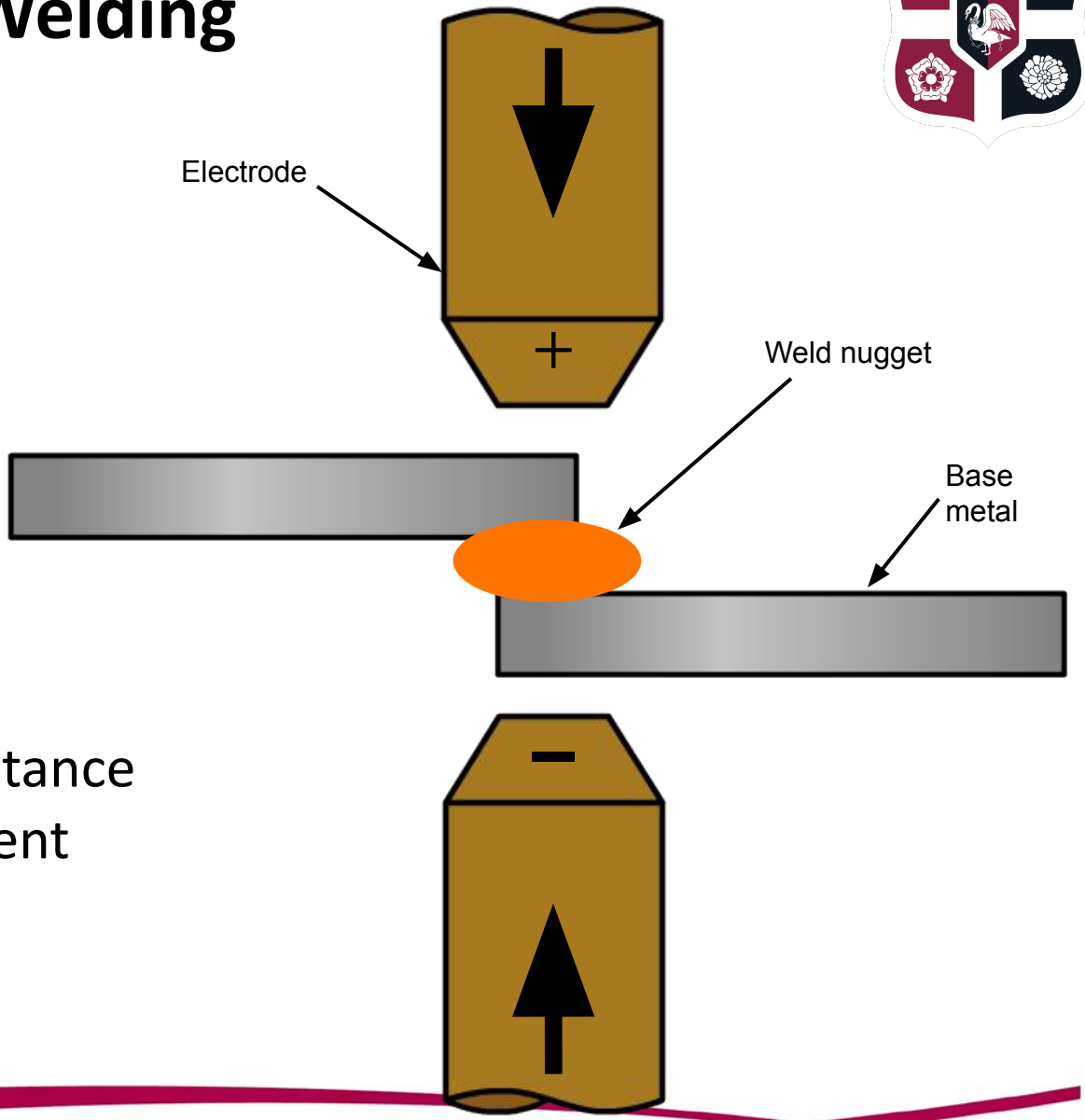


Resistance (Spot) Welding

Resistance (Spot) welding uses both heat and pressure to join metals.

The electrodes are forced together to generate pressure.

Heat is generated by resistance to the high electrical current flow at the welding point.



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Resistance (Spot) Welding



Types of Welding Joint

There are four common types of welding joint:

- Butt Joint



- Lap Joint



- T Joint



- Corner Joint



Types of Weld

There are two main types of weld:

- ***Groove welding*** – where the edge of the metal is machined to accept the weld



- ***Fillet welding*** – the edge of the metal is not machined for welding

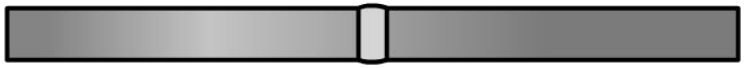


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Groove Welds

The type of groove weld depends upon the metal thickness:

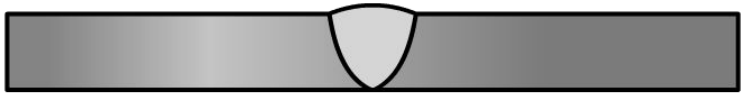
- *Square groove*



Square groove weld

Metal thickness
1.5 - 5 mm

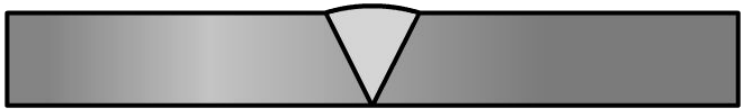
- *U groove*



U groove weld

Metal thickness
5 - 9 mm

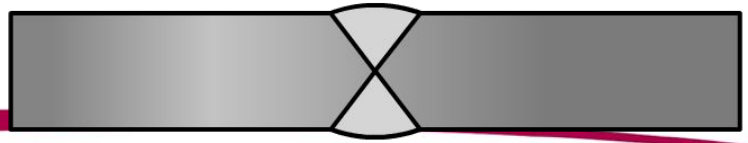
- *V groove*



V groove weld

Metal thickness
9 - 12 mm

- *Double-V groove*



Double-V groove weld

Metal thickness
>12 mm

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Fillet Welds



T-Joints



Lap joints



Corner joints

Fillet welds are ideal for T-joints, lap joints and corner joints.

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Question 2

Where are groove welds typically used?

A) In butt joints with prepared edges

B) In lap joints with unprepared edges

C) In T-joints with unprepared edges

D) In corner joints with unprepared edges

Next >

Question 2

Where are groove welds typically used?

A) In butt joints with prepared edges

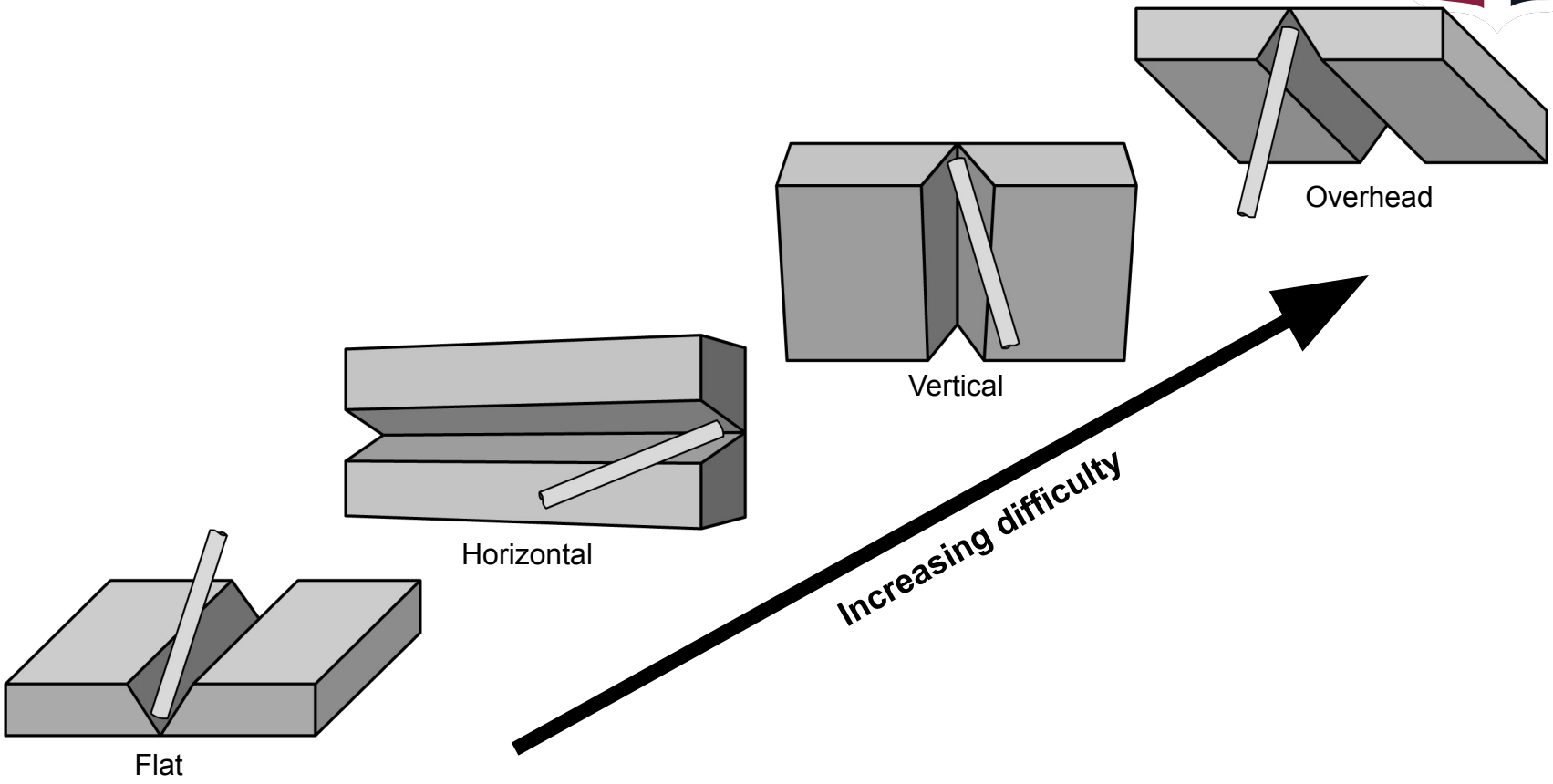
B) In lap joints with unprepared edges

C) In T-joints with unprepared edges

D) In corner joints with unprepared edges

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Welding Position



The orientation during welding is referred to as the welding position.

Next >

Question 3

Which position is the most difficult to weld in?

A) Flat

B) Horizontal

C) Vertical

D) Overhead

Next >

Question 3

Which position is the most difficult to weld in?

A) Flat

B) Horizontal

C) Vertical



D) Overhead

Next >

Welding Safety

Here are some typical welding hazards:

- *Fumes and gases*
- *Fire and explosion*
- *Heat and electric shock*
- *High intensity light from arcs*



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Advantages of Welding

Welding has
these advantages:

- *It is portable*
- *It is permanent*
- *It is usually stronger than the base metal*
- *It is economical in its use of materials*









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Disadvantages of Welding

Welding has these disadvantages:

- *Labour costs are high*
- *Disassembly is difficult*
- *It is difficult to see weld defects*
- *Heat causes thermal stress and changes the properties of the base metals*









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Summary

In this presentation you have seen:

- *different types of welding process*
- *different types of welding joint*
- *different types of weld*

End